STATISTICAL LEARNING THEORY (03/03/26)

Summer Term 2026

Other course names:

Machine Learning (Data Science Master)
Adv. Machine Learning (AIS Master)

MOTIVATION

Example:

	$\in \mathbb{R}$ Weight (in g)	$\in \mathbb{R}$ Color (in $[0,1]$)	$\in \{0,1\}$ Tasty?
Papaya 1	800	0.1	0
⋮	⋮	⋮	⋮
Papaya N	1200	0.7	1

0 for TASTY? means "not tasty" and 1 means "tasty".

Lets call the information in this table our training data.

Based on that, we try to find

$$\boxed{h: \mathbb{R} \times \mathbb{R} \to \{0, 1\}}$$

i.e., a function that takes a tuple of (weight, color) and outputs a prediction of whether a papaya is tasty (1) or not (0).

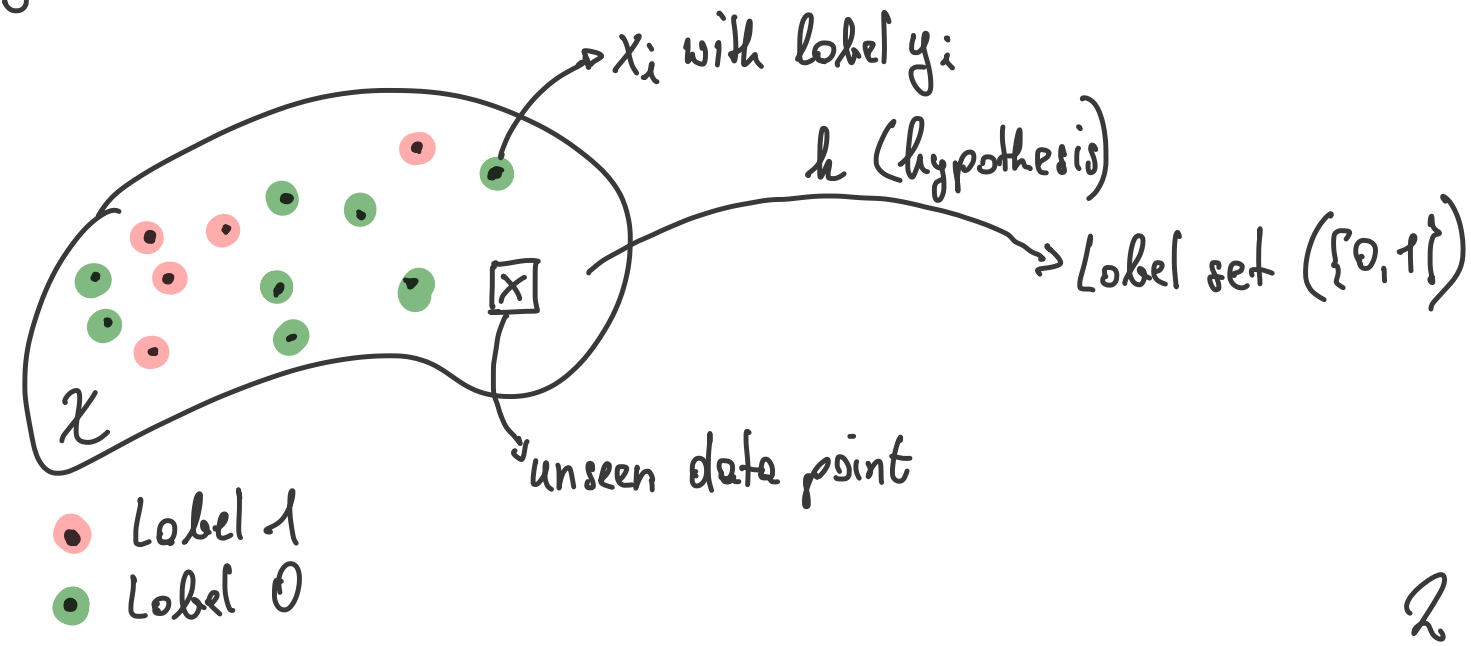
We are going to call such a function (h) a **hypothesis**.

More formally, the available data comes in the form of

$$((x_1, y_1), \ldots\ldots, (x_N, y_N)) = S$$

with (in our example) $x_i \in \underbrace{\mathbb{R}^2}_{\text{domain } \mathcal{X}}$ and $y_i \in \underbrace{\{0, 1\}}_{\text{Label set } \mathcal{Y}}$.

We say a **learner** (some algorithm) receives the training data S and returns h!



Two assumptions we are going to make initially:

A.1. All the x_i's are drawn <u>independently and identically distributed</u> (iid) from some (unknown) distribution D over our domain $\mathcal{X}$.

A.2. The x_i's are labeled by some (unknown) function

$$\boxed{f : \mathcal{X} \to \mathcal{Y}}$$

called the true labeling function.

This means our training data is of the form

$$S = \left((x_1, \underbrace{f(x_1)}_{y_1}), \ldots\ldots, (x_N, \underbrace{f(x_N)}_{y_N}) \right)$$

<u>What do we care about?</u>

true labeling function

$$\{ x \in \mathcal{X} : h(x) \neq f(x) \} = A$$

hypothesis

i.e., all points in our domain $\mathcal{X}$ for which h <u>differs</u> from f.

Specifically, we are interested in the probability of making a mistake $(h(x) \neq f(x))$

$$D(A) = D\left(\{ x \in \mathcal{X} : h(x) \neq f(x) \} \right) = \mathbb{P}_{x \sim D}\left[h(x) \neq f(x) \right]$$

$$(*) \quad \boxed{=: L_{D,f}(h)}$$ generalization error

The <u>empirical</u> version of (*) is

$$\frac{1}{N} \cdot \left| \left\{ i \in \{1, \dots, N\} : h(x_i) \neq f(x_i) \right\} \right| =: \boxed{L_S(h)}$$

We call this the empirical error/risk.

(<u>Notation</u>: $[N] = \{1, \dots, N\}$
$S|_x = (x_1, \dots, x_N)$)

We can also write the empirical error as

$$L_S(h) = \frac{1}{N} \cdot \sum_{i=1}^{N} \mathbb{1}_{h(x_i) \neq f(x_i)} \quad \text{with}$$

$$\mathbb{1}_{h(x) \neq f(x)} = \begin{cases} 1, & \text{if } h(x) \neq f(x) \\ 0, & \text{else} \end{cases}$$

<u>Claim</u> (relating L_S to $L_{D,f}$):

$$\mathop{\mathbb{E}}_{S|_x \sim D^N} \left[L_S(h) \right] = L_{D,f}(h)$$

<u>Proof:</u> next week!